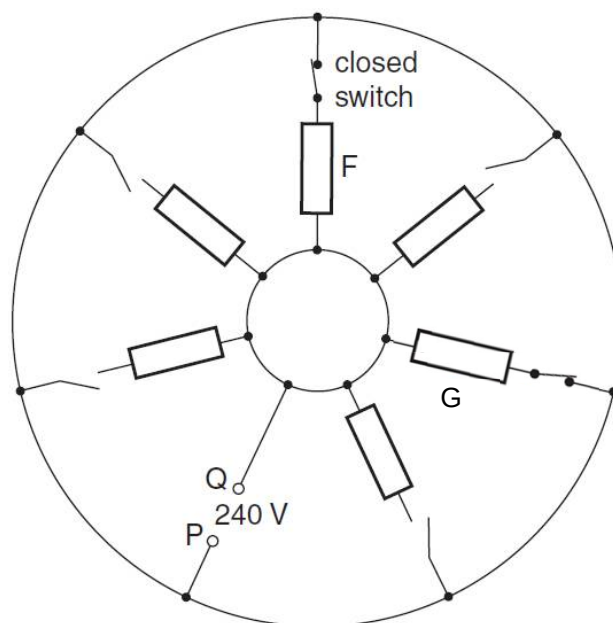


- 21** Electrical sockets in a house are connected to a circuit called a ring main. The circuit is connected between P and Q to the 240 V power supply as shown.

[Turn over



Two devices, F and G, are currently switched on. They have resistances of $1200\ \Omega$ and $1700\ \Omega$ respectively.

What is the current supplied by the power supply and total power dissipated by both devices?

	current / A	total power dissipated / W
A	0.083	20
B	0.083	82
C	0.34	20
D	0.34	82

Ans: D

F and G are connected in parallel to the power supply.

$$R_{\text{effective}} = \left(\frac{1}{1200} + \frac{1}{1700} \right)^{-1} = 703\ \Omega$$

$$I = \frac{240}{703} = 0.34\ \text{A}$$

$$P_{\text{total}} = \frac{240^2}{1200} + \frac{240^2}{1700} = 82\ \text{W}$$

22. A wire of length 2.0 cm is placed in the plane of the paper, along a line 60° clockwise from the